Tian (Phil) Xu

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Education

The Chinese University of Hong Kong **Sept 2026 - Sept 2030**
PhD in Finance Hong Kong SAR

- Research Interest: Empirical Asset Pricing, Statistical Modeling, and Machine Learning

Imperial College London **Aug 2024 - Aug 2025**
MSc in Risk Management and Financial Engineering London, UK

- **Relevant Modules**: Computational Finance with C++ (94%); Systematic Trading with Machine Learning (94%); Risk Management and Valuation (83.6%); Financial Statistics (83.2%); Investments and Portfolio Management (83.2%)
- **Final Grade**: Distinction (80.85%, ranked 1st out of 200 students)
- **Academic Achievements**: Outstanding Student Prize; Dean's List; Numerical Finance Prize

Southwestern University of Finance and Economics **Sep 2018 - Jun 2022**
Bachelor of Economics, Finance Chengdu, China

- **Relevant Modules**: Portfolio Management (93%); Data Analysis (96%); Financial Data Analysis (96%)
- **GPA**: 3.7/5.0

University of California, Berkeley **Jul 2020 - Dec 2021**
Bachelor's-level exchange program Berkeley, USA

- **Relevant Modules**: Modern Statistical Prediction and Machine Learning (A); Numerical Analysis (A)
- **GPA**: 3.96/4.0

London School of Economics and Political Science **Jul 2019 - Aug 2019**
Summer School London, UK

- **Module**: FM250 Finance (A)
- **GPA**: 4.0/4.0

Research Experience

Visiting Researcher **Sep 2025 - Present**
Imperial Business School London, UK

- **Supervisor**: Prof. Robert Kosowski and Dr. Farouk Jivraj
- **Affiliation**: Imperial Centre of Excellence in Quantitative Finance
- Conducting research under faculty supervision on empirical asset pricing and systematic trading strategies, focusing on the intersection of factor-based and machine-learning approaches.
- Developing an empirical project that uses hedge fund position and short-selling data to examine how managers adjust factor exposures and rebalance long-short portfolios across different market environments.
- Collaborating with Dr. Zaici Li on research that separates monetary- and fiscal-driven influences on the yield curve, aiming to build a unified decomposition that captures both monetary expectations and fiscal-policy effects.
- Developing standardized empirical workflows and analytical tools to support large-scale research in asset pricing, portfolio construction, and performance evaluation.

Master's Thesis: Empirical Asset Pricing via Transfer Learning **Aug 2025**
Imperial Business School

- **Supervisor**: Prof. Andrea Buraschi
- This paper proposes a transfer learning framework that embeds no-arbitrage latent factor structures into deep neural networks, bridging the gap between structural asset pricing theory and modern predictive modeling. The approach pretrains networks on simulated data generated from a no-arbitrage latent factor model and subsequently fine-tuned them on Fama–French 49 industry portfolios with firm characteristics, ensuring theory-consistent initialization and robust empirical adaptation. Empirical results demonstrate that the framework delivers higher out-of-sample R^2 and superior long–short portfolio Sharpe ratios, while retaining theoretical coherence relative to both econometric (IPCA) and unconstrained deep learning benchmarks.

Professional Experience

Imperial Business School: Student Investment Fund **Sep 2024 - Present**
Head of Quantitative Research London, UK

- Engaging with faculty supervisors on trend-following strategy research, preparing formal presentations and refining empirical methodologies through iterative feedback, culminating in adoption within the Fund's portfolio design.
- Conducting independent research in empirical asset pricing with machine learning methods, developing predictor specifications and model evaluations; results are incorporated into the Fund's systematic investment strategies.
- Applying the Instrumented PCA model in a real-world investment setting to estimate latent factors and assess the structure and persistence of risk premia across assets; the results support the Fund's systematic factor strategies.
- Implementing standardized empirical workflows by integrating Bloomberg, WRDS–CRSP, Compustat, and other academic datasets, and developing Python tools for data access, portfolio evaluation, and backtesting; together these tools support large-scale empirical asset-pricing analysis across research teams.
- Designing and leading training sessions on Python programming, quantitative databases, and empirical asset pricing methods, equipping new members with essential research skills.

AQUMON **Jun 2022 - Mar 2024**
Quantitative Researcher Hong Kong SAR

- Developed an AI-macro model, applying PCA on over 1,000 indicators to generate real-time GDP and CPI nowcasts; leveraged outputs for systematic timing strategies across equity and bond indexes, achieving a Sharpe ratio of 2.54.
- Developed a portfolio optimization framework, implementing a broad range of approaches, from classical mean–variance optimization and Black–Litterman to modern risk-based and downside-risk methods (risk parity, CVaR, regularized portfolios); established as the firm-wide standard and applied by institutional clients in portfolio construction.
- Extended the firm's risk attribution framework through tensor-based regression, providing a rigorous method to map portfolio exposures to 16 systematic factors, improving interpretability relative to traditional regressions.
- Constructed a comprehensive database of over 800,000 global mutual funds, integrating Refinitiv and Morningstar data with automated updates, supporting empirical research on fund evaluation and risk modeling.
- Led the launch and management of five quantitative asset allocation and fixed income funds with AUM exceeding CNY 200 million, achieving an annualized Sharpe ratio of 2.14-2.63 since inception.

AQUMON **Feb 2021 - Jun 2022**
Quantitative Researcher Intern Hong Kong SAR

- Contributed to the development of SmartFund, a factor-based mutual fund evaluation framework, applying asset pricing theories to assess funds across five dimensions (fundamentals, performance, manager, company, and risk); supported the construction of empirical scoring models deployed for over 20 institutional clients.
- Conducted volatility research in U.S. and Chinese equity markets, applying GARCH-type models and option-based hedging strategies; reduced max drawdowns from 6.5% to 2.3% (U.S.) and from 4.5% to 1.5% (China) in backtests.
- Conducted research on overfitting in financial predictive models, proposing a framework that applied ensemble methods, dropout, and large-scale parameter tuning; demonstrated improved robustness across diverse market conditions.
- Supported the research and development of SmartAdvice, a rule-based robo-advisory system translating asset allocation theory into systematic rebalancing rules; assessed strategy performance through backtesting and portfolio analysis.

China Securities Co., Ltd **Aug 2020 - Oct 2020**
Quantitative Investment Intern Beijing, China

- Conducted empirical research on factor models, replicating and testing a large set of published equity factors on Chinese market data to evaluate their predictive performance.

Awards & Scholarships

- **Outstanding Student** – Imperial College London, 2025 (awarded to the best MSc student for overall performance)
- **Dean's List** – Imperial College Business School, 2025 (awarded for academic excellence in the top 10% of students)
- **Numerical Finance Prize** – Imperial College London, 2025 (achieved the highest score in numerical finance modules)
- **Distinction in All MSc Modules** – Imperial College London, 2025 (achieved distinction across all nine courses)
- **Third Prize Scholarship** – Southwestern University of Finance and Economics, 2020 Fall, 2021 Spring and 2021 Fall
- **Honourable Mention** – Mathematical Contest in Modeling (MCM/ICM), 2019

Teaching Experience

Teaching Assistant for Intro to Asset Management — Nov 2025 - Dec 2025

Imperial Business School

- Delivered a lecture on empirical research data and infrastructure in asset management, introducing students to the use of financial databases and standardized workflows for academic and applied research.
- Supported the course by assisting in the preparation of teaching materials and providing guidance to students on course content and assignments.

Tutor for Financial Data Analysis and Programming — Sep 2021 - Apr 2022

Southwestern University of Finance and Economics

- Supported the following year's cohort with course projects, problem sets, and Python programming.

Technical Skills

- **Programming**: Proficient in Python (TensorFlow, scikit-learn), C++, Matlab, R, SQL; familiar with Java, Lisp.
- **Financial Databases**: Bloomberg, WRDS (CRSP, Compustat), Refinitiv, Morningstar; MySQL, SQLAlchemy.
- **Other Tools**: LaTeX, Linux, Git, GitHub, cloud computing and containers (Docker).